Week 1 Report

1. Data Source Profiles & Architecture

Yahoo Finance Ecosystem

While Yahoo Finance no longer provides an official public API, the data remains highly accessible through open-source wrappers that interact with Yahoo's internal endpoints.

- **Data Structure:** The architecture is centered around the **Ticker**—a unique identifier for a publicly traded asset.
- **Available Metrics:**
  - **Time-Series Market Data:** Historical Open, High, Low, Close (OHLC) pricing and trading volume.
  - **Fundamentals:** Financial statements, P/E ratios, and market capitalization.
  - **Derivatives:** Options chain data (calls, puts, strike prices, and expiration dates).
- **Limitations:** As an unofficial source, it is susceptible to rate limiting and sudden structural changes. It is best utilized for data exploration, rapid prototyping, and backtesting rather than live, production-grade systems.

UN Comtrade API

The UN Comtrade database is the definitive, official source for global import and export statistics, offering a highly standardized system for macroeconomic data retrieval.

- **Data Structure:** Access is strictly standardized using a RESTful API and specific classification systems rather than plain text searches:
  - **M49 Codes:** Numeric identifiers used for reporting and partner countries.
  - **HS Codes (Harmonized System):** Hierarchical classifications for traded goods, scaling from broad market sectors (2-digit) to highly specific commodities (6-digit).
- **Available Metrics:** Annual and monthly trade flows (Imports, Exports, Re-exports) quantified in dollar values and physical weights.
- **Limitations:** The free public tier restricts query volume. Large-scale historical aggregation across multiple nations generally requires premium access.

2. Conceptual Data Workflows

Extracting data from these sources requires standardizing the requests into structured, tabular formats (such as DataFrames) for downstream processing.

- **Market Data Retrieval:** Analysts typically initialize a connection to a specific asset via its ticker. The workflow involves requesting a specific temporal period and frequency (e.g., daily data over five years), which returns a clean time-series matrix ready for technical analysis or feature engineering.
- **Trade Data Retrieval:** Fetching macroeconomic data involves constructing a parameterized query containing the reporter country, partner country, trade direction, and commodity code. The resulting JSON payload from the UN server is then parsed into a structured table, linking trade values to specific global events or timeframes.